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Agentic FedGrid: Autonomous Privacy-Preserving Intrusion Detection via Federated Learning and Agentic AI

FINAL YEAR PROJECT WORK (BIS786)

SYNOPSIS

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Agentic FedGrid: Autonomous Privacy-Preserving Intrusion Detection via Federated Learning and Agentic AI

1. Introduction

Brief background of the topic: The proliferation of Internet of Things (IoT) devices, edge computing, and cloud-native architectures has led to a massive increase in heterogeneous network traffic. Historically, securing these vast digital ecosystems required centralizing all network logs and telemetry data into a single cloud repository to train machine learning-based Intrusion Detection Systems (IDS). While this centralized approach allowed for the creation of robust security models, it inherently exposed sensitive network behaviour to transit risks and required immense computational power.

Relevance of the problem: As cyber threats become increasingly sophisticated—manifesting as zero-day vulnerabilities, polymorphic malware, and Advanced Persistent Threats (APTs)—static, rule-based firewalls have proven inadequate. Simultaneously, strict data privacy regulations (such as HIPAA and GDPR) legally restrict organizations from sharing raw, localized network traffic. The modern cybersecurity landscape is trapped in a paradox: creating smarter AI defense systems requires massive amounts of data, but transmitting and storing that data centrally is a severe privacy and security liability.

Importance of the study: To resolve this bottleneck, there is an urgent need to transition toward decentralized, proactive defense mechanisms. By bringing the AI directly to the edge data—rather than transmitting the data to the AI—organizations can neutralize threats locally in real-time. This study is critical because it engineers an architecture that not only preserves absolute data privacy but also empowers edge nodes with autonomous decision-making capabilities, removing the latency and cost of centralized cloud processing.

Current scenario in industry/academia: In academia, Federated Learning (FL) is heavily researched as the premier privacy-preserving training method. However, existing FL deployments are fundamentally static; they require centralized servers to dictate when and how local edge devices should train. The cybersecurity industry is currently demanding a shift toward "Agentic AI"—systems that do not just passively learn, but possess the autonomy to hunt threats, detect behavioral shifts in network traffic, and trigger their own learning cycles without human intervention. This project bridges that exact gap between academic federated algorithms and industrial agentic autonomy.

2. Literature Review

1. Muthuswamy et al. (2026) - Hybrid Transformer and XGBoost Model for Federated IoT Intrusion Detection

- **Key Findings:** Proposed a hybrid architecture utilizing a Transformer backbone and an XGBoost tabular specialist in a Federated Learning environment. They successfully used Low-Rank Adaptation (LoRA) to share only lightweight adapter updates, reducing communication overhead by 89.5%.
- **Drawbacks:** The model struggled with complex, low-frequency web and injection attacks (such as SQL Injection and XSS) because these payloads closely resemble legitimate traffic, causing lower precision.
- **How our project addresses it:** Agentic FedGrid relies strictly on deep Tabular Transformers equipped with self-attention mechanisms, which are inherently better at mapping the complex, sequential relationships of sophisticated injection payloads than static XGBoost trees.

2. Devi et al. (2025) - Vision Transformers-Enhanced Cybersecurity Framework for Intrusion Detection

- **Key Findings:** Introduced FAT-ViT, a framework that converts network traffic logs into two-dimensional visual representations. The system uses Vision Transformers to capture spatial correlations of attacks and incorporates Contrastive Adversarial Training to improve robustness.
- **Drawbacks:** Converting tabular network traffic into 2D images requires significant mandatory preprocessing. This introduces a computational bottleneck before the AI can even begin analyzing the data, which is inefficient for low-power edge devices.
- **How our project addresses it:** We eliminate this preprocessing bottleneck entirely. Agentic FedGrid uses Tabular Transformers that ingest raw network logs natively, bypassing the need for image conversion while preserving high-speed edge compute efficiency.

3. Vijetha (2026) - Agentic Intelligence for Unified Cyber Defense

- **Key Findings:** Developed AISAF, an autonomous framework that detects concept drift using lightweight behavioral monitoring (latent-space deviations and prediction-confidence). It uses an agentic meta-controller to selectively retrain only the affected submodules, achieving a 0.93 Drift Recovery Index.
- **Drawbacks:** The system relies on fixed temporal window segmentation (e.g., batches of 512 samples). This fixed batching can underperform in ultra-low-latency streams where malicious drift occurs at sub-millisecond timescales.

- **How our project addresses it:** Agentic FedGrid incorporates Agentic AI that autonomously triggers training based on real-time traffic anomalies (via tools like mitmproxy) rather than waiting for a fixed static window to fill up, allowing for immediate, proactive mitigation.

4. Panwar & Abdelrahman (2025) - Agentic AI in Cybersecurity Review

- **Key Findings:** Analyzed the paradigm shift from traditional, rule-based cybersecurity to autonomous Agentic AI. Highlighted core agentic capabilities including self-learning security models, intelligent threat hunting, and automated incident response without human intervention.
- **Drawbacks:** Autonomous systems face severe risks, including susceptibility to adversarial AI attacks (evasion techniques), AI biases, and the "black-box" problem where automated decisions lack explainability and transparency.
- **How our project addresses it:** We actively counter these risks in two ways. First, we use Contrastive Adversarial Defense to immunize the model against evasion tactics. Second, we integrate an Explainability Dashboard (using attention rollout) to highlight exactly which network features triggered an automated response, ensuring complete transparency.

5. Adam & Baroudi (2024) - Federated Learning for IoT: Applications, Trends, Taxonomy (Survey)

- **Key Findings:** Provided a comprehensive taxonomy of Federated Learning in IoT ecosystems. Identified non-IID data distributions, device heterogeneity, and severe communication bottlenecks as the primary challenges for decentralized security.
- **Drawbacks:** As a survey, it outlines high-level challenges but does not propose concrete architectural solutions for intrusion detection, nor does it address the integration of deep learning with tabular data. Furthermore, standard Federated Learning transmits full model weights, which crashes low-bandwidth networks.
- **How our project addresses it:** Agentic FedGrid moves beyond standard Federated Learning. By deploying the Federated Averaging (FedAvg) algorithm via a serverless AWS orchestrator, we ensure that only lightweight mathematical intelligence crosses the network, entirely solving the communication bottleneck highlighted in this research.

4. Problem Statement

The Issue: Organizations and individuals cannot legally or safely share sensitive, localized network traffic logs with centralized cloud servers to train global threat-detection models. Consequently, localized edge devices are left relying on static, outdated firewalls that are fundamentally incapable of recognizing the complex, sequential behaviors of modern zero-day attacks and Advanced Persistent Threats (APTs).

The Gap: While existing Federated Learning solutions partially address the privacy concern, they suffer from high communication overhead and lack autonomous execution capabilities. Edge nodes in current research remain completely dependent on central orchestration to initiate learning, preventing them from proactively adapting to localized "concept drift" and mitigating new threats in real-time.

5. Objectives of the Study

- **Objective 1: Autonomous Edge Defense Engineering:** To engineer and deploy decentralized Agentic AI worker nodes (using Python) capable of autonomously intercepting local network traffic, detecting sequential anomalies using Tabular Transformers, and executing real-time defensive actions without requiring cloud authorization.

- **Objective 2: Privacy-Preserving Collaborative Intelligence:** To implement a robust intelligence-sharing pipeline utilizing the Federated Averaging (FedAvg) algorithm, mathematically guaranteeing that only lightweight model weight updates—and absolutely zero raw user data—are transmitted across the internet.

- **Objective 3: Zero-Cost Serverless Orchestration:** To architect a highly scalable, distributed backend using entirely free-tier AWS serverless components (API Gateway, Lambda, S3) that functions strictly as a zero-knowledge orchestrator to aggregate global intelligence at zero compute cost.

- **Objective 4: Adversarial Robustness and Evasion Defense:** To fortify the local Transformer models against sophisticated evasion techniques and adversarial AI attacks by integrating Contrastive Adversarial Training into the local agent learning loop.

- **Objective 5: Transparent and Explainable AI Reasoning:** To establish trust and auditability in autonomous security decisions by developing an Explainability Dashboard that utilizes attention-rollout visualizations to trace exactly which network features triggered a defensive action

6. Scope of the Project

- **What the project will cover:** The scope encompasses the development of local Agentic Python nodes, the training of localized Tabular Transformer models, and the DevOps architecture of a serverless AWS backend. It includes the integration of a secure HTTPS pipeline for weight uploading and the deployment of a visual Explainability Dashboard to map AI decision reasoning.

- **Limitations:** The project operates under strict privacy constraints; therefore, centralized data storage is entirely out of scope. No raw network traffic or private logs will ever touch the cloud databases. Additionally, the system will be rigorously evaluated using benchmark cybersecurity datasets (such as UNSW-NB15) and simulated proxy traffic rather than

deployment on a live, commercial enterprise network. Custom hardware fabrication (e.g., custom PCBs) is excluded.

- **Target domain/industry/application:** The primary domains are Cybersecurity, Edge Computing, and Privacy-Preserving Machine Learning, with direct applications for IoT security infrastructures, corporate enterprise networks, and decentralized threat intelligence sharing.

7. Methodology

Architecture and Workflow Diagram: The system is structured as a fully decentralized Edge-to-Cloud pipeline designed for zero-knowledge intelligence sharing.

1. **Edge Interception:** Autonomous local Python agents intercept live network traffic at the device level.

2. **Local Inference & Training:** The agent evaluates the traffic using a localized Tabular Transformer. If concept drift or an anomaly is detected, the agent autonomously triggers a local training loop to update its model weights.

3. **Secure Transmission:** The agent extracts only the modified, lightweight parameters (via LoRA adapters) and pushes them through a secure HTTPS POST request to the cloud.

4. **Zero-Knowledge Orchestration:** A cloud-based API Gateway receives the weights and triggers a serverless aggregation function. The cloud merges all incoming agent weights into a new, optimized global master model without ever seeing the raw network data.

5. **Grid Synchronization:** The updated master model is saved to a cloud storage repository and immediately synchronized back down to all edge agents, fortifying the entire grid against the newly discovered threat.

Algorithms and Techniques:

- **Tabular Transformers:** Deployed at the edge for deep sequential pattern recognition. By utilizing multi-head self-attention mechanisms, the model can look back across time-series network flows to detect the hidden sequences of complex, multi-stage attacks.

- **Federated Averaging (FedAvg):** The core mathematical aggregation algorithm utilized by the cloud orchestrator to securely merge disparate matrix weights from heterogeneous edge devices into a singular, globally optimized model.

- **Contrastive Adversarial Training:** A proactive defensive technique where local agents generate and train against simulated adversarial payloads, maximizing the mathematical distance between normal and malicious traffic to prevent evasion tactics.

Comprehensive Technology Stack:

- **Edge Compute & AI Modeling:** * *Python* (Core agent logic and traffic handling)
    - *PyTorch* (Deep learning framework for building and training the Tabular Transformers)
    - *mitmproxy* (For programmatic, real-time local network traffic interception and parsing)
- **Cloud Orchestration (AWS Serverless Stack):** * *Amazon API Gateway* (To act as the secure, high-throughput bridge receiving weight updates)
    - *AWS Lambda* (To execute the FedAvg mathematical aggregation algorithm at zero idle compute cost)
    - *Amazon S3* (To serve as the secure repository for distributing the updated global master model)
- **Explainability Dashboard & Frontend:**
    - *React.js* (Utilizing modern declarative patterns to build a highly responsive, scalable user interface)
    - *Visualization Libraries* (e.g., Recharts or Plotly to render dynamic attention-rollout visualizations, allowing security analysts to transparently track exactly which network features triggered the AI's defensive actions in real-time)

Step-by-Step Execution Plan:

1. **Data Engineering:** Parse, clean, and preprocess the UNSW-NB15 benchmark dataset to establish baseline benign and malicious feature vectors.
2. **Edge Node Development:** Code the autonomous Python worker agents and embed the PyTorch Tabular Transformer for local inference and self-training.
3. **Cloud Infrastructure Provisioning:** Architect the AWS serverless environment, writing the specific Python Lambda functions required to execute the FedAvg logic.
4. **Pipeline Integration:** Connect the edge agents to the AWS cloud environment, successfully establishing the secure parameter-sharing loop and synchronization process.
5. **Dashboard Integration & Testing:** Develop the React-based explainability dashboard, conduct adversarial stress testing via proxy traffic, and validate system resilience.

Evaluation Metrics: The framework will be rigorously evaluated against the following key metrics:

- **Security Efficacy:** Classification Accuracy (%), Precision, Recall, Macro F1-Score, and False Positive Reduction Rate.

- **Adaptation & Efficiency:** Drift Recovery Index (DRI) to measure how fast the agent recovers from zero-day concept drift, Inference Latency (ms/sample) to ensure real-time edge viability, and Cloud Compute Payload Size (KB) to prove communication efficiency.

8. Expected Outcomes

- **Expected performance improvements:** The system is expected to achieve enterprise-grade detection accuracy (>96%) against complex payloads and zero-day vulnerabilities. It will drastically reduce false positive rates through localized contextual learning and adversarial training, while maintaining a sub-50ms inference latency to ensure real-time edge protection.

- **Deliverables:** 1. A deployable Python software package representing the autonomous edge agent. 2. A fully configured, reproducible AWS serverless architecture pipeline. 3. The compiled, hybrid Tabular Transformer threat model. 4. An interactive Explainability Dashboard providing trace-logic for the model's predictions. 5. A comprehensive final engineering report and analysis documentation.

- **Academic contributions:** This project will provide a validated, tangible proof-of-concept demonstrating that advanced Agentic AI threat hunting and global intelligence aggregation can be achieved entirely at the network edge, eliminating massive cloud compute overhead while strictly adhering to data privacy protocols.

9. Work Plan / Timeline

Phase	Task	Duration
1	Literature review, finalization of the technological stack, and acquisition/preprocessing of the UNSW-NB15 dataset.	Week 1–2
2	Problem identification, system architecture flow mapping.	Week 3–4
3	Core Implementation: Coding the Python edge agents, PyTorch Tabular Transformers, and AWS Lambda FedAvg aggregation logic.	Week 5–8
4	Testing & Evaluation: Running simulated traffic via mitmproxy, conducting adversarial stress tests, and building the Explainability Dashboard.	Week 9-10
5	Documentation: Compiling performance metrics, generating the final project report, and preparing the jury presentation slides.	Week 11-12

10. References

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